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National Disaster Management Division · North Eastern Regional
Landslide Early Warning System

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SOFTWARE & TECHNICAL DESIGN DOCUMENT (SDD / TDD)

AEGIS-LEWS v2.0: System
Architecture & Engineering Blueprint

Mathematical Geotechnical Models, 2.4 GHz BLE Mesh Protocols, and
Sovereign Edge Cyberinfrastructure

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Author / Lead	AEGIS Core Architecture & RF Engineering Team	Reviewed / Approved By	Chief Technical Directorate, National Disaster Management Cyberinfrastructure
Security Clearance	NIC-CERT-IN Level-4 · Restricted Official Distribution	Target Environments	NER 8 States EOC Command & Citizen P2P Mesh

National Informatics Centre (NIC) · Disaster Management Cyberinfrastructure

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1. Architectural Overview & 4-Tier Decomposition

The AEGIS-LEWS platform is architected as a sovereign, four-tier hybrid computing system designed to guarantee continuous operational availability from high-altitude orbital sensors down to battery-powered citizen smartphones in disconnected Himalayan ravines:

Layer	Component	Technologies	Primary Operational Function
Tier 1: Space & In-Situ	Earth Observation & Hydro-Meteorological	Copernicus Sentinel-1 InSAR, Open-Meteo, IMD Doppler Radar	Line-of-sight displacement velocities (v_{LOS}) and orographic precipitation ingestion.
Tier 2: Cloud Engine	Geotechnical Computation & State Command	FastAPI, Uvicorn, NumPy/SciPy, Render Cloud Container	Infinite slope stability calculations, 18-sector hazard scoring, WebSocket telemetry bus.
Tier 3: Resilient Bus	Emergency Dispatch & Gateway Mesh	WebSockets (RFC 6455), Public Emergency Mesh (ntfy.sh), MQTT	Zero-latency dispatch fan-out to active gateway phones and regional EOC command centers.
Tier 4: Sovereign Edge	Autonomous Citizen Node & Hardware Siren	Android Capacitor, Java BLE Mesh, AudioTrack PCM, Local Flash Cache	Zero-internet 2.4 GHz BLE peer-to-peer relay, GPS safety radar, lockscreen emergency takeover.

2. Physics-Informed Geotechnical Model & Risk Mathematics

Unlike naive machine-learning models that behave as unexplainable black boxes, AEGIS-LEWS couples deterministic geomechanical limit-equilibrium equations with satellite and atmospheric inputs:

2.1 Infinite Slope Stability & Dynamic Factor of Safety

For translational bedrock landslides characteristic of the Northeast Himalayas, the dynamic Factor of Safety (FoS) is formulated as:

FoS = [$c' + (\gamma \cdot z - \gamma_w \cdot h_w) \cdot \cos^2 \beta \cdot \tan \varphi'$] / [$\gamma \cdot z \cdot \sin \beta \cdot \cos \beta + F_{seismic}$]

Where:

- c' = Effective soil/rock cohesion (kPa)
- γ = Moist unit weight of colluvial overburden (kN/m³)
- γ_w = Unit weight of water (9.81 kN/m³)
- z = Regolith slip plane depth (meters)
- h_w = Dynamic perched groundwater table height driven by cumulative rainfall (meters)
- β = Local terrain slope angle (degrees, derived from 30m SRTM DEM)
- φ' = Effective angle of internal shearing resistance (degrees)
- $F_{seismic}$ = Pseudo-static seismic acceleration force ($k_h \cdot \gamma \cdot z$)

2.2 InSAR Displacement Velocity Coupling

Satellite interferometric line-of-sight velocity (v_{LOS} , mm/day) from Sentinel-1 radar constellations provides ground-truth kinematics of active slope creeping:

$$\text{Shear_Strain_Rate } (\epsilon') = |v_{\text{LOS}}| / z_{\text{shear}}$$

$$\text{Failure_Probability} = 1.0 - \left(1.0 / \left(1.0 + \exp(-k \cdot (1.0 - \text{FoS}) + \lambda \cdot |v_{\text{LOS}}|) \right) \right)$$

When cumulative precipitation drives h_w to z and satellite interferometry registers an accelerated creep rate ($v_{\text{LOS}} > 25$ mm/day), the failure probability breaches **95%**, automatically tripping the emergency broadcast pipeline.

3. Bluetooth LE Mesh Network Architecture & RF Protocol

To ensure communication survivability during total infrastructure collapse, AEGIS-LEWS implements a custom sovereign broadcast protocol over standard 2.4 GHz BLE advertising physical channels (Channels 37, 38, and 39):

3.1 Dedicated Sovereign Service & Characteristic UUIDs

- **Sovereign Mesh Service UUID:** 0000AE61-0000-1000-8000-00805F9B34FB (Allocated for AEGIS emergency signaling)
- **Full Directive GATT Characteristic UUID:** 0000AE62-0000-1000-8000-00805F9B34FB (Readable by connecting rescue devices)

3.2 Custom 20-Byte Binary Protocol Data Unit (PDU) Layout

Because standard BLE advertising packets are hard-limited to 31 bytes by legacy Bluetooth specifications, the emergency broadcast is packed into an ultra-dense 20-byte binary frame injected into `ServiceData`:

Byte Index	Field Name	Data Type	Description & Bit Allocation
[0]	Magic Byte	UInt8	Always 0xAE (Identifies legitimate AEGIS sovereign emergency frame).
[1]	Packet Type	UInt8	0x01 : SOS 0x02 : Warning 0x03 : Beacon 0x04 : WebRelay.
[2]	Risk Score	UInt8	0 to 100 percentage integer representing failure probability.
[3..4]	Node ID Hash	Int16 (Big-Endian)	Deterministic hash of transmitting device ID (suppresses self-echoes).
[5..8]	Latitude	Int32 (Big-Endian)	WGS-84 coordinate encoded as Lat × 100,000 (precision ±1.1 meter).
[9..12]	Longitude	Int32 (Big-Endian)	WGS-84 coordinate encoded as Lon × 100,000 (precision ±1.1 meter).
[13]	Packed Meta	UInt8	Upper 4 bits: Severity Code (1=Safe..4=Critical) Lower 4 bits: Preset Code (1-6).
[14..19]	Text Snippet	Byte[6] UTF-8	Truncated zone name or directive summary for legacy non-GATT devices.

3.3 Autonomous Multi-Hop Flood Routing

Valley Flood Algorithm

1. A citizen device receives an emergency frame on BLE channel 37/38/39.
2. The packet is decoded; if Magic = 0xAE and NodeHash ≠ CurrentDeviceHash , processing begins.
3. If Severity ≥ HIGH , the native siren sounds and the device immediately schedules a **low-latency advertising cycle**.
4. The device re-broadcasts the identical payload with an incremented hop counter for 45 seconds.
5. This propagates the alarm down the mountain gorge from vehicle to vehicle over distances exceeding 2.5 km.

4. Native Android Subsystem Design

The Android native integration layer (aegis.lews.v2) bridges the Capacitor web application to low-level Linux kernel and Android framework APIs:

4.1 AegisSirenService (Persistent Foreground Engine)

- Runs as an un-killable Android Foreground Service typed as dataSync | connectedDevice (Android 14+ compliant).
- Maintains a PowerManager.PARTIAL_WAKE_LOCK ensuring CPU execution persists even when the device enters deep Doze mode.
- Directly controls AudioTrack in streaming mode, generating mathematical sine/sawtooth sound oscillations (750 Hz ↔ 1250 Hz) at 44.1 kHz sample rate to avoid reliance on external media decoders.

4.2 AegisBleMeshManager (Hardware Radio Controller)

- Binds to `BluetoothLeScanner` and `BluetoothLeAdvertiser` using Android 12+ permissions with `neverForLocation` assertion.
- Maintains an in-memory `ConcurrentHashMap` of discovered peers, dynamically computing relative distance using the Log-Distance Path Loss model: $\text{Distance} = 10^{((\text{TxPower} - \text{RSSI}) / (20 \cdot n))}$.
- Exposes high-speed asynchronous bindings to the React webview via the `@JavascriptInterface` `AegisBleBridge`.

5. System Sequence Workflows

5.1 Critical Breach Alert Dispatch Workflow

```

[Geotechnical Ingestion Engine]
|
| (1) Compute FoS = 0.67 (Breach Detected in Tawang Corridor)
v
[FastAPI Backend (/alerts/ble-broadcast)]
|
+----> [WebSocket Broadcast] ----> [EOC Authority Portal: Red Alert Banner]
|
+----> [ntfy.sh Emergency Bus] → [Internet Connected Gateway Phones]
|
| v
| [AegisBleMeshManager]
| |
| | (2) Start BLE Advertising (0xAE61)
| | v
| | [2.4 GHz Physical Radio RF Burst]
| | |
| | | v
| | | [Offline Citizen Phone (Zero SIM/Wi-Fi)]
| | | |
| | | | (3) Intercept Frame (Magic: 0xAE)
| | | | v
| | | | [Full-Screen Siren Wakeup & Local Mesh Relay]

```